Numerical Linear Algebra
Lecture 4: Vector Norms, Matrix Norms

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Vector Norms

Definition of Norms

- Norm captures "size" of vector or "distance" between vectors
- There are many different measures for "sizes" but a norm must satisfy some requirements:

Definition

A norm is a function $||\cdot||$: $\mathbb{C}^m \to \mathbb{R}$ that assigns a real-valued length to each vector. It must satisfy the following conditions:

1. $\|\mathbf{x}\| \geq 0$, and $\|\mathbf{x}\| = 0$ only if $\mathbf{x} = 0$
2. $\|\mathbf{x}+\mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$
3. $\|\alpha\mathbf{x}\| = |\alpha|\,\|\mathbf{x}\|$.

- An example is Euclidean length (i.e.$||\mathbf{x}|| = \sqrt{\sum_{i=1}^{m} |x_i|^2}$)

p-norms

- Euclidean length is a special case of p-norms, defined as

$$\|\mathbf{x}\|_p = (\sum_{i=1}^{m} |x_i|^p)^{\frac{1}{p}}$$

for $1 \leq p \leq \infty$

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- 1-norm: $\|\mathbf{x}\|_1 = \sum_{i=1}^{m} |x_i|$
- ∞-norm: $\|\mathbf{x}\|_\infty$. What is its value? Answer: $\|\mathbf{x}\| |_\infty = \max_{1 \leq i \leq m} |x_i|$
- Why we require $p \geq 1$? What happens if $0 \leq p < 1$?

Weighted p-norms

- A generalization of p-norm is weighted p-norm, which assigns different weights (priorities) to different components.
  - It is anisotropic instead of isotropic
- Algebraically, $\|\mathbf{x}\|_{\mathbf{W}} = \|\mathbf{W}\mathbf{x}\|$, where $\mathbf{W}$ is diagonal matrix with $i-$th diagonal entry $w_i \neq 0$ being weight for ith component
- In other words,

$$\|\mathbf{x}\|_{\mathbf{W}} = (\sum_{i=1}^{m} |w_i x_i|^p)^{\frac{1}{p}}$$

- What happens if we allow $w_i = 0$?

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- What happens if we allow $w_i = 0$?
- Can we further generalize it to allow $\mathbf{W}$ being arbitrary matrix?
- No. But we can allow $\mathbf{W}$ to be arbitrary nonsingular matrix.

Matrix Norms

Matrix Norms Induced by Vector Norms

- Viewing $m \times n$ matrix as mn-vectors is not always useful, as operations involving $m \times n$ matrices do not behave this way
- *Induced matrix norms* capture such behavior

Definition

Given vector norms $||\cdot||_{(n)}$ and $||\cdot||_{(m)}$ on domain and range of $\mathbf{A} \in \mathbb{C}^{m \times n}$, respectively, the induced matrix norm $||\mathbf{A}||_{(m,n)}$ is the smallest number $\mathbb{C} \in \mathbb{R}$ for which the following inequality holds for all $\mathbf{x} \in \mathbb{C}^n$:

$$\|\mathbf{A}\mathbf{x}\|_{(m)} \leq C \|x\|_{(n)} .$$

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- In other words, it is supremum of ratio $||\mathbf{A}\mathbf{x}||_{(m)}/||\mathbf{x}||_{(n)}$ for all nonzero vectors $\mathbf{x} \in \mathbb{C}^n$
- Maximum factor by which $\mathbf{A}$ can "stretch" $\mathbf{x} \in \mathbb{C}^n$

$$\|\mathbf{A}\|_{(m,n)} = \sup_{\mathbf{x} \in \mathbb{C}^n, \mathbf{x} \neq 0} \|\mathbf{A}\mathbf{x}\|_{(m)} / \|\mathbf{x}\|_{(n)} = \sup_{\mathbf{x} \in \mathbb{C}^n, \|\mathbf{x}\|_{(n)}=1} \|\mathbf{A}\mathbf{x}\|_{(m)}$$

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- Is vector norm consistent with matrix norm of $m \times 1$-matrix?

1-norm

- By definition

$$||\mathbf{A}||_1 = \sup_{\mathbf{x}\in\mathbb{C}^n, ||\mathbf{x}||_1=1} ||\mathbf{A}\mathbf{x}||_1$$

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- What is it equal to?
  - Maximum of 1-norm of column vectors of **A**
  - "maximum column sum" of **A** is oversimplified in the textbook
- To show it, note that for $\mathbf{x} \in \mathbb{C}^n$ and $||\mathbf{x}||_1 = 1$

$$||\mathbf{A}\mathbf{x}||_1 = ||\sum_{j=1}^{n} x_j \mathbf{a}_j||_1 \leq \max_{1\leq j\leq n} ||a_j||_1 ||\mathbf{x}||_1$$

- Let $k = \arg\max_{1\leq j\leq n} ||\mathbf{a}_j||_1$, then $||\mathbf{A}e_k||_1 = ||\mathbf{a}_k||_1$, so $\max_{1\leq j\leq n} ||a_j||_1$ is tight upper bound

$\infty-$norm

- By definition

$$||\mathbf{A}||_\infty = \sup_{\mathbf{x}\in\mathbb{C}^n, ||\mathbf{x}||_\infty=1} ||\mathbf{A}\mathbf{x}||_\infty$$

- What is $||\mathbf{A}||_\infty$ equal to?

$\infty-$norm

- By definition
$$||\mathbf{A}||_\infty = \sup_{\mathbf{x}\in\mathbb{C}^n, ||\mathbf{x}||_\infty=1} ||\mathbf{A}\mathbf{x}||_\infty$$
- What is $||\mathbf{A}||_\infty$ equal to?
  - Maximum of 1-norm of column vectors of $\mathbf{A}^\mathbf{T}$
- To show it, note that for$\mathbf{x} \in \mathbb{C}^n$ and $||\mathbf{x}||_\infty = 1$
$$||\mathbf{A}\mathbf{x}||_\infty = \max_{1\leq i\leq m}|\mathbf{a}_i^* x| \leq \max_{1\leq i\leq m}||\mathbf{a}_i^*||_1||\mathbf{x}||_\infty$$
where $\mathbf{a}_i^*$ denotes the $i-$th row vector of $\mathbf{A}$
- Furthermore, $\max_{1\leq i\leq m}||\mathbf{a}_i^*||_1$ is a tight bound.
  - Which vector can we choose to reach the bound?

2-norm

- What is 2-norm of a matrix?

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- Answer: Its largest singular value.
- We will talk more about singular-value decomposition

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- What is 2-norm of a diagonal matrix?

Cauchy-Schwarz and Holder Inequalities

- Holder inequality: Let p and q satisfy $1/p + 1/q = 1$ with $1 \leq p,\ q \leq \infty$, then

$$|\mathbf{x}^*\mathbf{y}| \leq ||\mathbf{x}||_p ||\mathbf{y}||_q$$

- Cauchy-Schwarz inequality

$$|\mathbf{x}^*\mathbf{y}| \leq ||\mathbf{x}||_2 ||\mathbf{y}||_2$$

- Cauchy-Schwarz inequality is a special case of Holder inequality

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- Cauchy-Schwarz inequality is a special case of Holder inequality
- Example: What is 2-norm of rank-one matrix? Hint: Use Cauchy-Schwarz inequality.

Bounding Matrix-Matrix Multiplication

- Let $\mathbf{A}$ be an $l \times m$ matrix and $\mathbf{B}$ an $m \times n$ matrix, then for $\mathbf{x} \in \mathbb{C}^n$

$$||\mathbf{AB}||_{(l,n)} \leq ||\mathbf{A}||_{(l,m)}||\mathbf{B}||_{(m,n)}$$

- To show it, note

$$||\mathbf{ABx}||_{(l)} \leq ||\mathbf{A}||_{(l,m)}||\mathbf{Bx}||_{(m)} \leq ||\mathbf{A}||_{(l,m)}||\mathbf{B}||_{(m,n)}||\mathbf{x}||_{(n)},$$

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- In general, this inequality is not an equality
- In particular, $||\mathbf{A}^n|| \leq ||\mathbf{A}||^n$ but $||\mathbf{A}^n|| \neq ||\mathbf{A}||^n$ in general for $n \geq 2$

General Matrix Norms

One can view $m \times n$ matrices as mn-dimensional vectors and obtain *general matrix norms*, which satisfy (for $\mathbf{A}$, $\mathbf{B} \in \mathbb{C}^{m \times n}$)

1. $||\mathbf{A}|| \geq 0$, and $||\mathbf{A}|| = 0$ only if $\mathbf{A} = 0$
2. $||\mathbf{A} + \mathbf{B}|| \leq ||\mathbf{A}|| + ||\mathbf{B}||$
3. $||\alpha \mathbf{A}|| = |\alpha| ||\mathbf{A}||$

Frobenius Norm

- One useful norm is Frobenius norm (a.k.a. Hilbert-Schmidt norm)

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n}|a_{ij}|^2} = \sqrt{\sum_{j=1}^{n}\|\mathbf{a}_j\|_2^2}$$

  i.e., 2-norm of mn-vector

- Furthermore,

$$\|\mathbf{A}\|_F = \sqrt{tr(\mathbf{A}^\mathbf{T}\mathbf{A})}$$

  where $tr(\mathbf{B})$ denotes trace of B, the sum of its diagonal entries

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- Furthermore,

$$\|\mathbf{A}\|_F = \sqrt{tr(\mathbf{A}^\mathbf{T}\mathbf{A})}$$

  where $tr(\mathbf{B})$ denotes trace of B, the sum of its diagonal entries
- Note that

$$\|\mathbf{AB}\|_F \leq \|\mathbf{A}\|_F \|\mathbf{B}\|_F$$

  because

$$\|\mathbf{AB}\|_F^2 = \sum_{i=1}^{n}\sum_{j=1}^{m}|\mathbf{a}_i^*\mathbf{b}_j|^2 \leq \sum_{i=1}^{n}\sum_{j=1}^{m}(\|\mathbf{a}_i^*\|_2 \|\mathbf{b}_j\|_2)^2 = \|\mathbf{A}\|_F^2 \|\mathbf{B}\|_F^2$$

Invariance under Unitary Multiplication

Theorem

For any $\mathbf{A} \in \mathbb{C}^{m \times n}$ and unitary $\mathbf{Q} \in \mathbb{C}^{m \times m}$, we have

$$\|\mathbf{QA}\|_2 = \|\mathbf{A}\|_2 \text{ and } \|\mathbf{QA}\|_F = \|A\|_F$$

In other words, 2-norm and Frobenius norms are invariant under unitary multiplication.
Proof for 2-norm: $\|\mathbf{Qy}\|_2 = \|\mathbf{y}\|_2$ for $\mathbf{y} \in \mathbb{C}^m$ and therefore $\|\mathbf{QAx}\|_2 = \|\mathbf{Ax}\|_2$ for $\mathbf{x} \in \mathbb{C}^n$.
It then follows from definition of 2-norm.